Clinical Decision Algorithm for Diabetes Management

CBME Clinical Skills - Flowchart-Based Decision Making

EMERGENCY DIABETES MANAGEMENT ALGORITHM

Step 1: Initial Assessment - ABCDE Approach

🚨 **AIRWAY (Airway)**:

- Is patient able to speak? ✅ → Proceed to Breathing
- No or poorly → Immediate airway management + Call for help

🫁 **BREATHING (Breathing)**:

- Normal respiratory rate (12-20/min) & pattern? ✅ → Proceed to Circulation
- Abnormal (Kussmaul breathing, acetone smell)? → Immediate IV fluids + Insulin protocol

💓 **CIRCULATION (Circulation)**:

- BP normal? Pulse regular? Skin normal? ✅ → Proceed to Disability
- Hypotensive/shock? → Emergency IVF + vasopressors if available

🧠 **DISABILITY (Disability)**:

- Alert & oriented (GCS 15)? ✅ → Proceed to Exposure
- GCS <15? Altered sensorium? → Check blood glucose immediately

👀 **EXPOSURE (Exposure)**:

- Remove clothes to check for infection/injection sites
- Blood glucose monitoring with glucometer
- Vital signs documentation

DIABETES DIAGNOSIS FLOWCHART

When to Suspect Diabetes

Start → High Suspicion →

```
Polyuria + Polydipsia
OR
Fatigue + Unexplained wt loss
OR
Recurrent infections
OR
Random BG > 200 mg/dL
```

```
OR
Positive risk factors
```

Yes → Proceed to Diagnostic Tests No → Consider Other Conditions

Diagnostic Pathway

```
Patient Presents with Symptoms
        ↓
Perform Random Blood Glucose
        ↓
RBS ≥ 200 mg/dL + Symptoms?
        ↓
Yes → DIABETES DIAGNOSED
No → Perform OGTT or HbA1c
        ↓
FPG ≥ 126 mg/dL or 2hr ≥ 200 mg/dL or HbA1c ≥ 6.5%
        ↓
Yes → DIABETES DIAGNOSED
No → PREDIABETES (IGT/IFG) or NORMAL
```

DIABETES MANAGEMENT ALGORITHM

Newly Diagnosed Type 2 Diabetes

```
Confirm Diagnosis
        ↓
Assess Risk Factors & Complications
        ↓
Patient Profile Assessment
├── Age, Duration, CVD Status
├── BMI, Motivation, Economic Status
└── Hypoglycemia Risk, Co-morbidities
        ↓
Treatment Selection
├── Lifestyle Only (Diet + Exercise)
├── Monotherapy (Metformin preferred)
├── Dual Therapy (Add DPP4-i or SGLT2-i)
├── Triple Therapy (Insulin + 2 OHA)
└── Insulin Only (If severe symptoms)
        ↓
Individualize Targets
├── HbA1c < 6.5% (Young, no CV disease)
├── HbA1c < 7.0% (Most patients)
├── HbA1c 7.0-8.0% (Elderly, comorbidities)
└── HbA1c < 8.5% (Limited life expectancy)
```

Hypoglycemia Management Algorithm

```
Blood Glucose < 70 mg/dL Detected
        ↓
Patient Conscious and Able to Swallow?
        ↓
Yes → Rule of 15 Protocol
├── Give 15g Fast-acting Carbohydrates
├── Wait 15 minutes, Recheck BG
├── If still <70 mg/dL, repeat 15g carbs
├── When BG >70 mg/dL, eat regular meal
└── Document cause and adjust treatment
        ↓
No/Glucagon Required → IM/SQ Glucagon 1mg
├── Recheck BG every 10-15 min
├── When conscious, give oral carbs
├── Hospitalization if no improvement
└── Teach prevention strategies
```

COMPLICATIONS SCREENING ALGORITHM

Annual Complications Screening

```
Diabetes Duration >1 Year OR New Diagnosis
        ↓
Perform All 5 Screening Tests
├── 1. Retinopathy: Fundoscopy/dilated exam
├── 2. Nephropathy: Urine ACR, eGFR
├── 3. Neuropathy: Foot exam, monofilament
├── 4. CVD: ECG, lipid profile, BP
└── 5. Foot CARE: Vascular assessment
        ↓
Any Abnormal Finding?
        ↓
Yes → Urgent Referral + Management
├── Severe retinopathy → Ophthalmologist (Monthly)
├── CKD Stage 3-5 → Nephrologist (3-6 months)
├── Foot ulcers → Surgeon/General Physician
├── Ischemic heart disease → Cardiologist
└── Uncontrolled risk factors → Optimization
        ↓
No → Continue Routine Care + Annual Screening
```

Diabetes Foot Ulcer Algorithm

```
High-Risk Foot Identified
        ↓
Daily Self-Inspection at Home
        ↓
Ulcer / Infection Signs Present?
        ↓
Yes → Non-Weight Bearing + Antibiotics
├── Daily dressing (Saline + Povidone-iodine)
├── Elevation + Rest for affected foot
├── Oral antibiotics (Amoxicillin-clavulanate)
├── Review in 48-72 hours
├── Education on prevention
└── Referral if deteriorates
        ↓
No → Continue Prevention
├── Proper footwear education
├── Foot care practices
├── Orthotic devices if needed
├── Regular podiatry visits
└── Control of risk factors (BG, smoking, lipids)
```

REFERRAL DECISION TREE

Primary Care to Secondary/Tertiary Level

```
Stable Diabetes Controlled at PHC?
        ↓
Yes → Continue PHC Follow-up
├── Monthly SMBG and BP recording
├── Quarterly HbA1c monitoring
├── Annual complication screening
├── Lifestyle reinforcement
└── Medication compliance monitoring
        ↓
No → Assess Referral Priority
├── Urgent Issues → Immediate Referral (<24 hrs)
│   ├── Suspected DKA/HHS
│   ├── Severe hypoglycemia (persistent)
│   ├── Foot ulcers with systemic features
│   ├── New-onset chest pain
│   └── Severe hyperglycemia (>400 mg/dL) symptomatic
│
├── Semi-Urgent → Referral within 1-2 weeks
│   ├── Uncontrolled BG (HbA1c >9%)
│   ├── New complications identified
│   ├── Medication side effects requiring change
│   └── Pregnancy planning
│
└── Routine → Referral within 2-4 weeks
```

```
    ├── Specialist consultation for optimization
    ├── Young patients with early complications
    ├── Multiple comorbidities requiring coordination
    └── Research participation consideration
```

COMMUNITY INTERVENTION ALGORITHMS

Screening Camp Flowchart

```
Camp Preparation (1 week prior)
├── Community mobilization (ASHA/posters/loudspeakers)
├── Equipment check and calibration
├── Team training and role allocation
├── Consent form preparation
└── Emergency response planning
        ↓
Registration & Triage
├── Token system for patient flow
├── Basic demographics collection
├── Risk factor quick assessment
├── Aggravation to laboratory station
└── Separate areas for sensitive patients
        ↓
Laboratory Testing Stage
├── Quality check of glucometer strips
├── Consent verification and explanation
├── Blood glucose testing (finger prick)
├── Result interpretation and counseling
├── Referral slip provision if needed
└── Community health education
        ↓
Results Counseling & Follow-up
├── Normal results: Prevention advice + recall
├── Pre-diabetes: Lifestyle counseling + 3-month review
├── Diabetes: Immediate care + medication initiation
├── Complication signs: Urgent specialist referral
└── Next appointment scheduling
        ↓
Data Analysis & Quality Assurance
├── Camp data compilation and analysis
├── Identified gaps and community needs
├── Quality improvement suggestions
└── Report preparation for district authorities
```

PATIENT EDUCATION SUPPORT ALGORITHMS

Lifestyle Modification Counseling

```
Patient Readiness Assessment
        ↓
Motivational Interviewing
├── Explore personal reasons for change
├── Assess confidence and barriers
├── Set realistic goals
└── Build support systems
        ↓
Specific Goals Setting
├── Weight: BMI reduction 1-2 kg/month
├── Exercise: 150 min/week moderate activity
├── Diet: 45-65% carbs, <7% saturated fats
├── Smoking cessation program referral
└── Stress management techniques
        ↓
Follow-up and Reinforcement
├── Weekly check-ins for progress
├── Barrier identification and solutions
├── Family involvement strategies
├── Positive reinforcement and celebration
└── Long-term maintenance planning
```

Self-Monitoring Education Pathway

```
Patient Receives Glucometer
        ↓
Device Operation Training
├── Assembly and battery insertion
├── Test strip insertion method
├── Finger site selection and cleaning
├── Blood volume drop application
└── Result interpretation and recording
        ↓
SMBG Schedule Development
├── Newly diagnosed: 2-3 times daily
├── Insulin users: Pre/post meals + bedtime
├── Oral medications: Weekly monitoring
├── Stable disease: 2-3 times weekly
└── Hypoglycemia prone: More frequent testing
        ↓
Data Interpretation and Action
├── Trend analysis with healthcare provider
├── Adjustment of medications based on patterns
├── Recognition of hypo/hyperglycemia
├── Emergency thresholds and responses
└── Sharing data during clinic visits
```

IMPLEMENTATION NOTES

Algorithm Customization

- **Adapt to local resources**: What equipment and medications are available in your setting?
- **Cultural considerations**: Religious dietary restrictions, family involvement preferences
- **Resource constraints**: When PHC level vs. referral indicated due to limited local capacity
- **Cost considerations**: Generic medications vs. brand choices, transportation for referrals

Quality Assurance

- **Algorithm compliance**: Audit 10% of cases for protocol adherence
- **Outcome monitoring**: Track referral completion rates and patient follow-up
- **Continuous improvement**: Update algorithms based on new evidence and local experience
- **Training requirements**: All healthcare providers using algorithms should be trained

Documentation Requirements

- **Decision recording**: Which branch of algorithm followed and why
- **Patient consent**: For all diagnostic and treatment decisions
- **Referral documentation**: Reason, urgency level, receiving facility
- **Outcome follow-up**: What happened to referred patients

Clinical Decision Support: These algorithms provide evidence-based pathways for common diabetes management scenarios. Adapt to local resources and always use clinical judgment for individual patient needs.

Faculty Guide: Use these algorithms during OSCE stations and case discussions to reinforce systematic clinical reasoning skills.